Lecture 6 Worksheet

Name: _____

1. Initializing weights

Slides 13–14: chalkboard

Fix Apartment A: $x = (1, 1, 1, 1)$. The entries of $W^1 \in \mathbb{R}^{4\times 16}$ are independent $\mathcal{N}(0, s_1^2)$ draws; $b^1 = 0$.

$$a^1 = xW^1, \quad h^1 = \mathrm{ReLU}(a^1), \quad q_i = \frac{\mathbb{E}\|h^i\|_2^2}{m_i}, \quad m_0 = 4,\ m_1 = 16, \quad q_0 = 1.$$

Follow the derivation: expand $(a_j^1)^2$, take expectations over the weights, account for ReLU, and choose s_1^2 so that $q_1 = q_0$.

2. Practice the next layer

Slide 15

Layer 2 has 16 incoming units. Use $q_2 = (16s_2^2/2)q_1$ and He variance $2/d$, where d is the number of incoming units. Complete the three blanks.

Starting weights	s_1^2	q_1 (given)	s_2^2	q_2
He initialization	$1/2$	1		
Variance 1 in both layers	1	2	1	

3. Backward pass

Slide 17: chalkboard

Here $x = h^1 \in \mathbb{R}^{1\times 16}$. Entries of $W^2 \in \mathbb{R}^{16\times 16}$ are independent $\mathcal{N}(0, s^2)$ draws, $s^2 = 2/16$, and $b^2 = 0$.

$$a = xW^2, \quad h = \mathrm{ReLU}(a), \quad g = \frac{\partial L}{\partial h}, \quad \delta = \frac{\partial L}{\partial a}, \quad u = \frac{\partial L}{\partial x}.$$

For this scale estimate, approximate g as independent of the ReLU gates and δ as independent of W^2. For nonzero x, each gate is active with probability $1/2$. Follow the derivation from $\delta_j = g_j \mathbf{1}_{\{a_j>0\}}$ and $u = \delta(W^2)^\mathsf{T}$ to a relation between $\mathbb{E}\|u\|_2^2$ and $\mathbb{E}\|g\|_2^2$.

4. Layer Normalization

Slide 19

For $h = (1, 3)$, compute the mean μ, variance v, and normalized output. Use $\gamma = (1, 1)$, $\beta = (0, 0)$, and $\epsilon = 0$.

$$\mu = \frac{1}{m}\sum_{j=1}^{m} h_j, \qquad v = \frac{1}{m}\sum_{j=1}^{m}(h_j - \mu)^2, \qquad \mathrm{LN}(h)_j = \gamma_j \frac{h_j - \mu}{\sqrt{v + \epsilon}} + \beta_j.$$

5. Input standardization

Slides 21–22

a. Let $\hat{x} = (x - \mu)/\sigma$, where μ and $\sigma > 0$ are fixed feature statistics. Find w' and b' such that $b' + w'\hat{x} = b + wx$ for every x.

b. Now use a centered scalar feature and no bias. The data are fixed, and v denotes the weight for the standardized input.

$$\frac{1}{N}\sum_{i=1}^{N} x_i = 0, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N} x_i^2 > 0, \qquad \hat{x}_i = \frac{x_i}{\sigma}.$$

Compute the first and second derivatives of both losses:

$$L(w) = \frac{1}{2N}\sum_{i=1}^{N}(wx_i - y_i)^2, \qquad \widetilde{L}(v) = \frac{1}{2N}\sum_{i=1}^{N}(v\hat{x}_i - y_i)^2.$$

$L'(w) =$ $\widetilde{L}'(v) =$

$L''(w) =$ $\widetilde{L}''(v) =$

6. Minibatch gradients

Slide 24

At fixed w, draw B examples independently with replacement. Each example has gradient $g_b = L'(w) + \xi_b$, with $\mathbb{E}[\xi_b] = 0$ and $\mathrm{Var}(\xi_b) = \sigma^2$.

$$\bar{g} = \frac{1}{B}\sum_{b=1}^{B} g_b = L'(w) + \varepsilon, \qquad \varepsilon = \frac{1}{B}\sum_{b=1}^{B} \xi_b.$$

Compute $\mathbb{E}[\varepsilon]$ and $\mathrm{Var}(\varepsilon)$. Use $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$ and additivity of variance for independent variables. Here σ^2 is gradient-noise variance.

7. Tracking the error with noisy gradients

Slide 26: board work

Use fixed η, $\lambda > 0$, and fresh noise ε_t independent of e_t, with mean zero and variance σ^2/B at every step.

$$L(w) = \frac{\lambda}{2}(w - w^\star)^2, \quad g_t = \lambda(w_t - w^\star) + \varepsilon_t, \quad w_{t+1} = w_t - \eta g_t,$$

$$e_t = w_t - w^\star, \qquad M_t = \mathbb{E}[e_t^2].$$

Follow the board work. Derive e_{t+1}, expand its square, and take expectations. Evaluate the cross term and the noise-square term to obtain the final M_{t+1} recurrence.